

diploma/certificate exams, quality approval and other development activities.

Together, there are the government-run universities, the IITs and IIMs, the RECs that are partly government-funded and the private institutions—all with different management and administration.

The other problem, which has come to the fore with Tamil Nadu's doing away with the Common Admission Test, is the different selection procedures for each of the institutions or group of institutions—so there is JEE for the IITs, CAT for the RECs, state level exams for the state level colleges, GATE for the MTechs and so on.

Challenges

"It is a myth that only the premium institutions are producing the best; there are many institutes doing a good job," asserts Arulmani, Head of HR and Administration, Igarashi Motors. However, there are miles to go before the institutes can rest. Shortage of staff, and high-quality staff at that, is the recurrent theme in the technical education scenario in India. There are not enough qualified teachers. The IITs produce 60 per cent of the MTechs and 75 per cent of the PhDs today. Better pay at the industry and the lure of the industrial experience, in addition to the famed brain drain to the western countries, are a common cause for concern. The other recent development, the boom in the ITes market and their need for graduates from any stream (employment has grown from 284,000 in 1999-2000 to 1 million in 2004-2005) has led to many bright students being pulled away to work instead of pursuit of education.

The government, too, since the Fourth Five-Year Plan, has shifted its focus away from technical education to elementary and adult education, with a result that only Rs 100 crore has been allocated for technical education.

A weak R&D culture, poor manpower planning, almost negligent linkage with the industry, poor participation of women and slow response to changes threaten to weaken the technical education scenario in the country.

Initiatives

The need of the hour is action. China is a real threat. A professor from Purdue University mentioned at an HR Summit conducted by NASSCOM recently that engineers coming from China are almost as good as those graduating from the IITs. And, unless there is uniformity across all colleges, the English language alone will not help India become a global destination for bagging contracts.

Recognising this, the NBA was created in 1987 to certify the quality of the institutions.

One of the problems faced by the industry today, is the lack of uniformity in technical education, as different institutions are run differently

Any institute that has functioned for five years can apply for this. The rate of accreditation in the country currently is low, says Nath, with just 200 of the existing 5,000 colleges having been given the status. Infrastructure and manpower limitations at the AICTE themselves are a major concern in this regard. "Of course, students prefer word of mouth and the past performance of the institutions before they apply." However, what also needs to be addressed, in this context, is the huge capitation fee charged by the institutes and not reported by students.

NASSCOM, recognising the lacunae in the Indian technical education system, has signed up with AICTE for a faculty development program in the IT sector, in partnership with the IT industry. Such initiatives are needed across streams and industries. Says Kiran Karnik, President, NASSCOM, "The industry can play a major role in ensuring that the curriculum is up-to-date and relevant by sharing trends, likely future developments and the requirement of skill-sets." As part of the memorandum of understanding (MoU) signed between the two, NASSCOM and AICTE will jointly undertake curriculum review, develop training modules, develop a database of engineering and management colleges across the country and study international models for industry-academia partnership.

Nath also suggests that for improving the communication skills of the students, it would be a good idea to follow the REC model of admitting students from rest of the country also. "A mix of students from different regions will mean more interactions, a broadening of perspectives and better communication skills," he suggests.

Retention: The Bane

The IT industry's retention problems are well known. But, this industry itself has added to the retention problem of other industries as the demand for functional experts is growing. With better pays, that is a lure difficult to overcome. But some of the retention tactics applicable to the IT industry are applicable here too: Creating a conducive work atmosphere, providing the scope to innovate, delegating responsibilities, changing personal success symbols from just monetary rewards to roles and responsibilities and retraining and enhancing skill sets.

In the modern world, no job is for life and loyalty to the employer has been replaced by loyalty to the designation or job role. Recognising this is of utmost importance and approaching HR problems in this perspective will be helpful in better attraction and retention of employees. ■



How Secure Are You?

The growing dependence of information technology in modern society has created new vulnerabilities. Breaches in information security can have grim consequences.

The spectrum of products and procedures aimed at providing information security is vast. One of the most common and useful technologies is encryption.

This report provides current information on encryption and other emerging technologies and trends driving the development of new information security products, applications, and markets.

Encryption provides a secure communication channel even when the underlying system and network infrastructure is not secure. This is important when data passes through shared systems or network segments where many people may have access to the information. In these situations, sensitive data and passwords should be encrypted in order to protect them from unintended disclosure or modification.

Key Factors Affecting The Encryption Marketplace

Many organisations have produced encryption products, which include software, services, servers, and consulting. The market segments vary from PC users, to financial institutions to

Encryption and other information security technologies are receiving heightened attention now, thereby driving the development of new information security products, applications, and markets

military and government. Several approaches to encryption have been developed but there are substantial barriers to widespread deployment. For example, there are many different and incompatible encryption techniques available, and not all the software needed implements a common approach. Effective use of encryption requires vendors to agree on methods and standards. Two different cryptographic methods are being applied to computer security problems: private-key and public-key encryption.

The general public is concerned about access to private information. In many countries, there is already strong and public debate about both data security and secrecy. Many experts say that security concerns of the average home computer user are unfounded, although the public at large has not been very well informed to date. Improvements in security provided by public-key cryptography and other encryption systems are actually already providing much higher security than existed before, and public education will go a long way to addressing this issue.

Relatively weak encryption appears to have been used to protect files recovered from two computers believed to have belonged to Al-Qaeda operatives in Afghanistan. The files were found on a laptop and desktop computer bought by Wall Street Journal reporters from